

LOGIC CIRCUIT DESIGN

3-Bit Synchronous Binary Counter

Digital Logic Design Project · 2024/2025

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1. Idea of Operation

A 3-bit synchronous binary counter is a sequential digital circuit that counts from 000 to 111 in binary, incrementing by one on each rising clock edge. It is built using three JK flip-flops that share the same clock signal, making all state transitions simultaneous — hence the term synchronous.

Operating Principle

Each flip-flop represents one bit of the 3-bit count: QA (LSB), QB, and QC (MSB). Their toggling behaviour is controlled as follows:

Flip-Flop	Bit	Toggle Condition
FF-A (QA)	LSB	Toggles on every clock pulse → $J_A = K_A = 1$
FF-B (QB)	Middle bit	Toggles when $Q_A = 1$ → $J_B = K_B = Q_A$
FF-C (QC)	MSB	Toggles when $Q_A \text{ AND } Q_B = 1$ → $J_C = K_C = Q_A \cdot Q_B$

The counter cycles through all 8 states (000 → 001 → 010 → ... → 111 → 000). Because all flip-flops are clocked simultaneously, there is no ripple delay, resulting in faster and more reliable operation compared to asynchronous (ripple) counters.

2. JK Flip-Flop Excitation Table

The excitation table defines the required J and K inputs to produce a specific state transition $Q_n \rightarrow Q_{n+1}$. The symbol X denotes a don't-care condition (the input value does not matter for that transition).

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Key observations:

- $0 \rightarrow 0$: J must be 0; K is irrelevant (X)
- $0 \rightarrow 1$: J must be 1; K is irrelevant (X)
- $1 \rightarrow 0$: K must be 1; J is irrelevant (X)
- $1 \rightarrow 1$: K must be 0; J is irrelevant (X)

3. Counter Truth Table

The complete truth table shows the current state (QC QB QA), the next state (QC* QB* QA*), and the required J and K inputs for each flip-flop derived using the excitation table above.

QC	QB	QA	QC*	QB*	QA*	JC	KC	JB	KB	JA	KA
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Note: X = don't-care condition derived from the JK flip-flop excitation table.

4. Karnaugh Maps (K-Maps)

K-maps simplify the Boolean expressions for each J and K input by grouping adjacent 1s and don't-cares (X). Green cells = 1 (include); yellow cells = X (don't-care, can be grouped); white cells = 0.

K-Map for JC

QC·QB \ QA	00	01	11	10
QA=0	0	X	X	X
QA=1	0	1	X	X

Grouping: Group the two 1-cells in column 01 (QB=0,QA=1) — they are adjacent. Both require QB=1 and QA=1.

$$\text{Result: } JC = QB \cdot QA$$

K-Map for KC

QC·QB \ QA	00	01	11	10
QA=0	X	X	X	X
QA=1	X	1	X	X

Grouping: All X except one 1 cell. The entire map simplifies to $QB \cdot QA$.

$$\text{Result: } KC = QB \cdot QA$$

K-Map for JB

QC·QB \ QA	00	01	11	10
QA=0	0	X	X	0
QA=1	1	X	X	1

Grouping: The two 1-cells (bottom-left and bottom-right corners) form a group. Only QA=1 in common.

$$\text{Result: } JB = QA$$

K-Map for KB

QC·QB \ QA	00	01	11	10
QA=0	x	0	0	x
QA=1	x	1	1	x

Grouping: Two 1-cells in the bottom row, columns 01 and 11. QA=1 is the common factor.

Result: $\mathbf{KB = QA}$

K-Map for JA

QC·QB \ QA	00	01	11	10
QA=0	1	1	1	1
QA=1	x	x	x	x

Grouping: All QA=0 cells are 1 (entire top row). Groups with all X in bottom row → constant 1.

Result: $\mathbf{JA = 1}$

K-Map for KA

QC·QB \ QA	00	01	11	10
QA=0	x	x	x	x
QA=1	1	1	1	1

Grouping: All QA=1 cells are 1 (entire bottom row) → constant 1.

Result: $\mathbf{KA = 1}$

5. Derived Boolean Expressions

After minimisation with Karnaugh maps, the input equations for each JK flip-flop are:

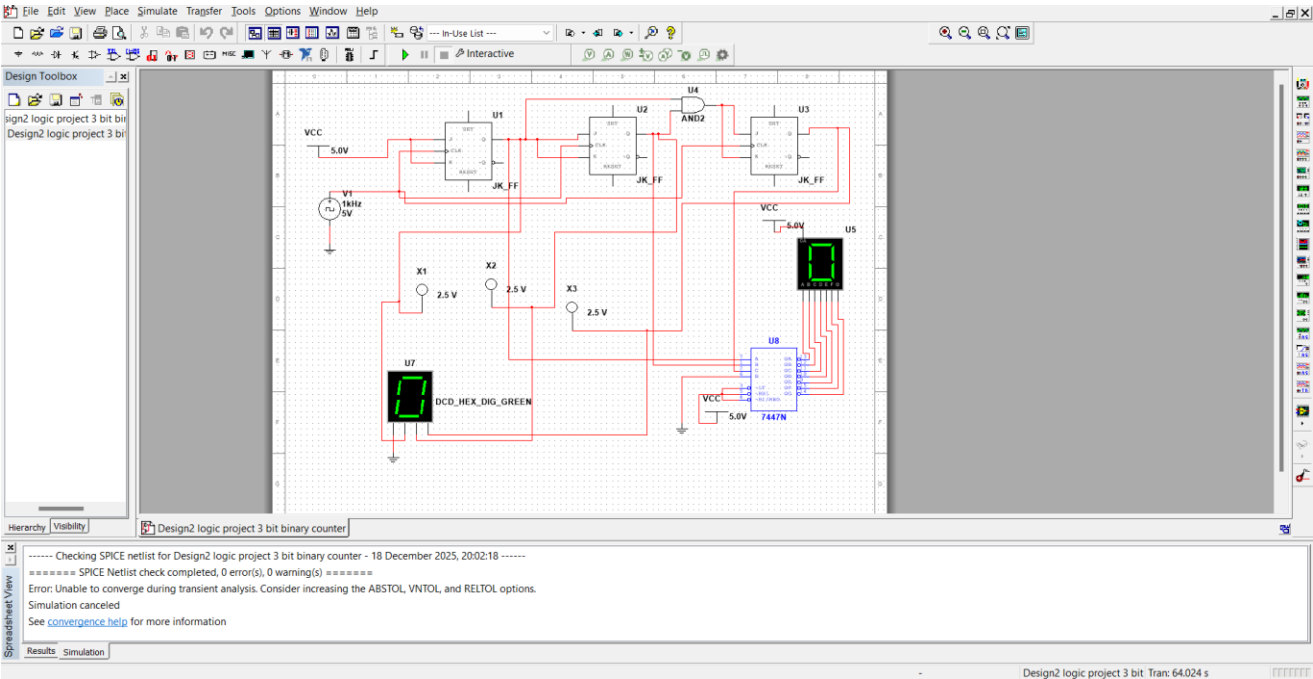
Flip-Flop	Input	Boolean Expression
FF-A (LSB)	$J_A = K_A$	1 (always toggle)
FF-B	$J_B = K_B$	Q_A
FF-C (MSB)	$J_C = K_C$	$Q_B \cdot Q_A$

These expressions confirm the design described in the Idea of Operation section: the AND gate connecting Q_A and Q_B feeds directly into J_C and K_C of the MSB flip-flop.

6. Simulation & Hardware

Software Simulation

The circuit was implemented and verified in a SPICE-based logic design environment. The schematic includes three JK flip-flop modules interconnected with an AND gate as derived from the Boolean expressions. A 7-segment display was connected to the output to show the count visually.



Simulation Parameter	Details
Software	SPICE-based digital logic simulator (Design2)
Components used	3× JK flip-flop, 1× AND gate, 1× 7-segment display, VCC (5 V)
Clock sources	3× independent 2.5 V pulse sources (X1, X2, X3)
Output	7-segment display (common-cathode, GREEN) showing count 0–7
Simulation mode	Interactive / Transient analysis
Result	Netlist check passed: 0 errors, 0 warnings

Hardware Implementation

The physical prototype was assembled on a solderless breadboard using standard 74-series ICs. The output was displayed on a 7-segment LED display. Two photos below show the breadboard at different count states, confirming correct hardware operation.

Hardware Parameter	Details
Platform	Solderless breadboard
Flip-flop ICs	74LS76 or equivalent dual JK flip-flop
Display	7-segment LED display (common-anode/cathode)
Power supply	External DC supply (~5 V)
Clock	Manual push-button or external oscillator
Status	Verified – display shows correct binary count

The breadboard photos demonstrate the working circuit at two different counter states, with the 7-segment display clearly showing the corresponding digit.